Project Title
Land Use Classification of Satellite Imagery using Transfer Learning with ResNet18

Project Category
Image Classification

Team Members
Catarina Silva (nº 23421)
João Alexandre (nº 29602)

Problem Statement
This project focuses on the automatic classification of land use and land cover from satellite imagery using deep learning. Accurate land-use mapping is essential for environmental monitoring, urban planning, agriculture and sustainable resource management. Traditional manual interpretation of satellite images is slow, costly and difficult to scale.
This work is relevant in the context of Green Data Science, as satellite imagery enables large-scale monitoring of environmental changes and supports data-driven decision-making in sustainability applications.

Challenges
A key challenge is the visual similarity between land-cover classes such as PermanentCrop, Pasture and HerbaceousVegetation, which share similar spectral and textural characteristics. In addition, satellite images show high variability due to seasonal changes, geography and lighting conditions, increasing classification difficulty. Ensuring good generalization and avoiding overfitting are also important challenges.

Dataset
The EuroSAT RGB dataset will be used, consisting of approximately 27,000 labelled Sentinel-2 satellite images across ten land-use classes, including AnnualCrop, Forest, Highway, Industrial, Residential, River and SeaLake.
The dataset is publicly available on Kaggle and will be split into training, validation and test sets. Data augmentation will be applied to the training set to improve variability and model robustness.

Method/Algorithm
The approach is based on Transfer Learning using a pre-trained ResNet18 model implemented in PyTorch. The network, originally trained on ImageNet, will be adapted to EuroSAT by replacing the final classification layer with a new layer of ten outputs.
Training will be performed in two stages: first, only the final layer is trained while freezing the convolutional backbone; second, the last residual block will be fine-tuned using a smaller learning rate to better adapt high-level features to satellite imagery.

Evaluation
Performance will be evaluated on a held-out test set using classification accuracy as the main metric. Precision, recall and F1-score will be computed per class for a more detailed analysis.
A confusion matrix will be used to identify misclassification patterns between similar classes, while training and validation curves will be analyzed to assess convergence and detect overfitting. These metrics will provide a comprehensive evaluation of the effectiveness of transfer learning for satellite image classification.